A Reduced-Order, Radially Coupled Model of the BYU Entrained-Flow Coal Gasifier: Formulation, Results, Validation, and Limitations

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***Abstract*—This paper documents the formulation, implementation, numerical checks, and experimental comparison of a reduced-order model for the atmospheric Brigham Young University (BYU) entrained-flow gasifier. The case uses Utah bituminous coal, oxygen-rich primary gas containing argon and steam, and a steam-only secondary stream. Two complementary calculations are used within one modeling workflow: a steady one-dimensional cross-section-averaged plug-flow calculation, and an axisymmetric annular calculation that couples reacting lanes through conservative radial species and enthalpy transport. The chemistry includes competing two-step coal devolatilization, element-balanced volatile allocation, three kinetic-diffusion char reactions, and a four-reaction global gas-phase set. The final radial mesh has 120 annuli and 600 axial stations. At that mesh the tracked C/H/O/N elemental-flow residual is $2.05 \times 10^{-6}\%$, the outlet area-average temperature is 3556.7 K, and the carbon conversion is 53.52%. The highest recovered temperature reaches the solver's 4000 K upper bracket and is therefore not a resolved physical peak. Dry inert-free outlet fractions, normalized over CO, $\mathrm{H_2}$, $\mathrm{CO_2}$, and $\mathrm{CH_4}$, are 58.66, 16.68, 24.21, and 0.45 vol%, respectively; the outlet also retains approximately 4.02 vol% dry area-average $\mathrm{O_2}$ on a total dry-gas basis. The measured BYU outlet values summarized by the reference study are 51.1% CO, 24.7% $\mathrm{H_2}$, 23.6% $\mathrm{CO_2}$, 0.3% $\mathrm{CH_4}$, and 82% carbon conversion. The radial calculation conserves the tracked elements and exhibits modest changes under grid refinement, but its outlet and profile errors show that numerical consistency is not equivalent to physical validation. The paper makes all spatial, chemical, thermal, and numerical assumptions explicit, maps the study outputs to the required project deliverables, and identifies model upgrades needed before predictive use.**

***Index Terms*—Coal gasification, entrained-flow reactor, reduced-order model, devolatilization, char conversion, radial mixing, validation, syngas.**

I. Introduction

Entrained-flow gasification converts pulverized coal and an oxidant into a hot product gas containing mainly CO and $\mathrm{H_2}$. In the reactor, particles heat rapidly, release volatile matter, form char, and undergo oxidation and gasification while the gas composition and temperature vary strongly in space. The BYU laboratory reactor is a useful benchmark because its geometry and feed conditions are documented and local gas samples are available for comparison. Brown *et al.* describe measured and predicted local and outlet behavior in this reactor [1]; Mularski and Modlinski later use the BYU case to examine how devolatilization, gas-phase chemistry, and char-conversion modeling affect predictions [2].

The present work has two aims. First, it turns the selected BYU case into a transparent executable model whose balances, kinetics, boundary conditions, and numerical choices can be inspected. Second, it evaluates the model against the benchmark quantities that matter for an engineering interpretation: feed description, axial species evolution, radial-axial species and temperature fields, outlet composition and conversion, mesh sensitivity, and comparison with measured profiles.

The model is deliberately described at the level it implements. It is a reduced-order reacting stream-tube model with radial diffusion coupling. It does not solve the Navier–Stokes equations, turbulent momentum transport, Lagrangian particle trajectories, radiative transfer, or a detailed chemical mechanism. The reference calculation in [2] is a two-dimensional axisymmetric CFD model with substantially more detailed turbulence, particle, radiation, and chemistry closures. Comparisons are consequently interpreted as a model assessment on the same experimental case, not as a claim that the present code reproduces the full reference CFD.

The main contributions and evidence are:

- a documented BYU case definition with coal properties, primary and secondary feeds, geometry, and particle-distribution assumptions;
- a steady axial reacting calculation and a radially coupled annular calculation with six requested species fields (CO, $\mathrm{H_2}$, $\mathrm{CO_2}$, $\mathrm{H_2O}$, $\mathrm{O_2}$, and $\mathrm{CH_4}$) and temperature;
- explicit chemical and numerical assumptions, including atom-balance closure, the hot-start energy addition, radial mixing closure, and temperature limits;
- grid-refinement and particle-size sensitivity results;
- centerline and radial comparisons with approximate experimental markers, with measurement and digitization uncertainty disclosed; and
- a deliverable-coverage map connecting the technical requirements to report sections, figures, and reproducible project artifacts.

II. Benchmark Problem and Required Output Coverage

A. Physical system

The modeled vessel is a straight cylindrical reactor of diameter $D = 0.20$ m and length $L = 1.80$ m, operated at 101.325 kPa. The selected coal is Utah bituminous coal.

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The published case supplies coal and gas rates, proximate and ultimate analyses, and a Rosin–Rammler particle-size distribution [2]. The two gas inlets are coaxial: the primary stream contains oxygen, argon, and steam and carries the coal; the secondary stream is steam. The input values used by the executable case file are listed in Table I.

B. Deliverable coverage

Table II gives a direct evidence map for the technical and presentation outputs. The six species fields are shown separately as axial profiles and as radial-axial contours. Results are supported by the saved numerical tables, the source code, the live-demonstration notebook, the class presentation, and this report.

III. Reference Study and Model Scope

The selected published study applies an axisymmetric CFD model to the BYU reactor, discretizes the domain with approximately 10^5 cells, and reports a grid-independence check. It uses a realizable k-ϵ turbulence model, a discrete-random-walk particle dispersion treatment, discrete-ordinates radiation with a weighted-sum-of-gray-gases model, competing two-step devolatilization rates tuned against FG-DVC, a global gas mechanism checked against detailed chemistry, and empirical kinetic-diffusion char conversion parameters optimized with CBK/E and CBK/G [2]. The study reports average and maximum local-profile errors for CO, H2, CO2, and H2O; it cautions that H2O values are derived by hydrogen balance and carry about $\pm 14\%$ uncertainty.

The present implementation reuses the same BYU geometry, feed case, published competing two-step devolatilization constants, and optimized char-kinetic parameters. It also uses the published nozzle dimensions from [3] to initialize radial inlet weights. These shared inputs make the cases comparable. The solver architecture and closure set differ substantially: chemistry is solved in axial stream tubes and then operator-split with radial diffusion; there is no solved velocity field, turbulence field, particle trajectory, radiation field, recirculation zone, or interphase momentum coupling. In addition, volatile composition and gas-phase rates are less detailed than in the reference model. The comparison is therefore a transparent reduced-model exercise with known differences in fidelity.

Two spatial formulations are used under the same case definition, not as sequential project phases. The cross-section-averaged plug-flow calculation is a simpler baseline and is used for the particle-size sweep. The radially coupled calculation generates the two-dimensional species and temperature fields and supplies the centerline and radial validation results.

IV. Mathematical Model

A. State variables and axial species balances

The axial calculation is steady. For gas species i, $F_i(z)$ is the molar flow rate in $\mathrm{mol\,s^{-1}}$. The solid state includes un-devolatilized dry-ash-free coal, $m_{\mathrm{raw}}(z)$, and carbon char, $m_{\mathrm{char}}(z)$, each in $\mathrm{kg\,s^{-1}}$. The generic axial species balance is

$$\frac{dF_i}{dz} = A_c \sum_{r=1}^{N_r} \nu_{i,r}\, \dot{\omega}_r + S_{i,\mathrm{het}} + S_{i,\mathrm{vol}}, \tag{1}$$

where $A_c = \pi D^2/4$, $\nu_{i,r}$ is the gas-phase stoichiometric coefficient, $\dot{\omega}_r$ is the global volumetric rate in $\mathrm{kmol\,m^{-3}\,s^{-1}}$, and the heterogeneous and volatile-release terms are converted to molar flow per unit axial length. Gas composition is computed from the species molar flows. The gas density and superficial velocity follow the ideal-gas relation at the fixed pressure:

$$\rho_g = \frac{P\overline{W}}{R_u T}, \qquad u_g = \frac{\dot{m}_g}{\rho_g A_c}. \tag{2}$$

The particles are assigned the gas velocity; slip, particle acceleration, and particle radial motion are not represented.

B. Energy balance and temperature recovery

Rather than integrate temperature as an independent state, the code recovers temperature at each axial state from a total enthalpy-flow constraint:

$$H(T, \mathbf{F}, m_{\mathrm{char}}, m_{\mathrm{raw}}, m_{\mathrm{ash}}) = H_0. \tag{3}$$

Species enthalpies include standard formation enthalpy and sensible enthalpy, with constant species heat capacities. Solid char, ash, and raw coal use constant heat-capacity approximations. A bracketed scalar root solve over 280–4000 K finds T; if the target is outside that enthalpy range, the state is returned at the nearer bracket endpoint. The reference state is set by the 1400 K hot-start patch, not by the cold feed enthalpy alone. Because reaction products carry their formation enthalpies, the constant-total-enthalpy closure converts reaction progress into sensible temperature without separately integrating a heat-release source.

This is an adiabatic closure. The wall boundary condition and radiative exchange are not solved. The single-temperature assumption sets particle temperature T_p equal to gas temperature T_g, although real coal particles can depart from gas temperature during rapid heating. Coal moisture is added immediately as vapor at the inlet; there is no separate liquid-droplet evaporation model. The clipping behavior means that a reported 4000 K point indicates the numerical upper bracket, not a validated prediction of a physical flame temperature.

The BYU case does not provide a calorific value in the selected input table. The code estimates the dry-ash-free higher heating value (HHV) from the ultimate analysis with the Channiwala–Parikh correlation [4]:

$$\begin{aligned}\mathrm{HHV_{daf}}\ [\mathrm{MJ\,kg^{-1}}] = {} & 0.3516C + 1.16225H - 0.1109O \\ & + 0.0628N + 0.10465S,\end{aligned} \tag{4}$$

where C, H, O, N, S are the dry-ash-free mass percentages. The resulting coal formation enthalpy is then estimated from the complete-combustion products and this HHV. This introduces a fuel-energy estimate rather than a measured calorific value.

C. Coal devolatilization and volatile allocation

Coal devolatilization uses the competing two-step model (C2SM). For the remaining raw coal, the two rates are

$$k_j(T_p) = A_j \exp\left(-\frac{E_j}{R_u T_p}\right), \quad j \in \{1, 2\}, \tag{5}$$

TABLE I
BYU CASE DEFINITION USED IN THE SIMULATION. VALUES MARKED AS ASSUMPTIONS ARE MODEL INPUTS RATHER THAN MEASURED VALUES TABULATED FOR THIS CASE.

Item	Value	Role in model
Reactor	$D = 0.20$ m; $L = 1.80$ m; $P = 101325$ Pa	Cylindrical, atmospheric case [2]
Coal	Utah bituminous; 23.88 kg/h	Fed with the primary stream [2]
Proximate analysis	Moisture 2.4 wt%; volatile matter 45.6 wt%; fixed carbon 43.7 wt%; ash 8.3 wt%	As-received basis [2]
Ultimate analysis	C 77.60, H 6.56, N 1.42, S 0.55, O 13.87 wt%	Dry ash-free basis; O is by difference [2]
Primary stream	26.24 kg/h at 400 K; 85.0% O_2, 12.6% Ar, 2.4% H_2O	Coal-bearing oxidant stream [2]
Secondary stream	6.62 kg/h at 473 K; pure steam	Coaxial steam stream [2]
Particle size	Rosin–Rammler; range 1–85 μm; mean 36 μm; spread 1.033	Published distribution [2]
Particle density	1300 kg/m^3	Provisional engineering assumption used to estimate particle count and surface area
Injector dimensions	Primary inner diameter 13 mm; secondary outer diameter 28.6 mm	Taken from the BYU setup reported by Abani and Ghoniem [3]; not tabulated in [2]
Inlet radial regularization	20% uniform-background contribution	Numerical assumption to avoid exactly empty/stagnant annuli
Wall and heat transfer	Adiabatic wall; gas and particle share one temperature	Reduced-model assumptions; wall heat loss and separate particle temperature are not solved
Hot-start temperature	1400 K at $z = 0$	Numerical ignition patch; its added energy is quantified in Section VII

TABLE II
REQUIRED OUTPUT COVERAGE AND EVIDENCE PRODUCED BY THE PROJECT.

Output	Where addressed in this paper	Reproducible artifact
Feed and major gas species	Section II-A; Table I; outlet composition in Section VI-B	Case YAML; inlet and outlet CSV files
Axial species distribution	Section VI-A; Fig. 1	Area-average and near-axis profile CSV; plot PNG
Species contours	Section VI-A; Fig. 2 for CO, H_2, CO_2, H_2O, O_2, and CH_4	Radial state and contour PNG files
Temperature distribution	Section VI-A; Fig. 3	Temperature field and area-average temperature CSV/PNG
Comparison with the selected journal study and BYU experiment	Sections VI-B–VI-D; Tables VI–VIII	Digitized marker CSV, comparison figures, and error tables
Numerical checks and sensitivity	Sections V and VI-E; Tables V and IX	Grid-study and particle-sweep CSV/PNG files
Conclusions and limitations	Sections VIII–IX	This report; source README
Report, code, and oral presentation	Source and packaging notes in Section XI	IEEE PDF and Overleaf source; Batch 18 source package; live-demo notebook; class presentation PDF

$$\dot{m}_{\mathrm{vol}} = (\alpha_1 k_1 + \alpha_2 k_2)\, m_{\mathrm{raw}}, \tag{6}$$

$$\dot{m}_{\mathrm{char}} = [(1-\alpha_1)k_1 + (1-\alpha_2)k_2]\, m_{\mathrm{raw}}. \tag{7}$$

The implementation uses $A_1 = 4.269 \times 10^4\,\mathrm{s}^{-1}$, $E_1 = 3.71 \times 10^7\,\mathrm{J\,kmol}^{-1}$, $\alpha_1 = 0.246$, $A_2 = 9.785 \times 10^6\,\mathrm{s}^{-1}$, $E_2 = 8.993 \times 10^7\,\mathrm{J\,kmol}^{-1}$, and $\alpha_2 = 0.931$, transcribed from Table 5 of [2]. The activation energies are converted to $\mathrm{J\,mol}^{-1}$ in the code before evaluating (5).

The mass released as volatiles does not determine a unique gas composition. Three elemental equations (C, H, O) cannot determine five product amounts (CO, CO_2, H_2, H_2O, and CH_4) without additional closure assumptions. The implementation therefore states two explicit allocation choices: all available volatile oxygen is first assigned to CO ($f_{\mathrm{O}\to\mathrm{CO}} = 1$); half the hydrogen remaining after water formation is allocated to methane ($f_{\mathrm{H}\to\mathrm{CH_4}} = 0.5$), with the rest assigned to hydrogen. Product allocations are clipped by the available C, H, and O inventories in a fixed sequence so they cannot exceed the elemental inventory.

Char is treated as pure carbon. Nitrogen is represented as N_2. Any volatile carbon left after CO, CO2, and methane allocation is retained as a soot/char surrogate and is subsequently consumed by the same char reaction set. This keeps the tracked C/H/O/N balances closed, but is not equivalent to the reference study's explicit tar species and tar chemistry. Sulfur is not represented by a gas species or included in

the tracked elemental-balance diagnostic; hence the reported balance check is specifically for C, H, O, and N, not for all elements in the fuel.

D. Heterogeneous char conversion

Three surface reactions are included:

$$\mathrm{C(s)} + \tfrac{1}{2}\mathrm{O_2} \rightarrow \mathrm{CO}, \tag{8}$$

$$\mathrm{C(s)} + \mathrm{CO_2} \rightarrow 2\mathrm{CO}, \tag{9}$$

$$\mathrm{C(s)} + \mathrm{H_2O} \rightarrow \mathrm{CO} + \mathrm{H_2}. \tag{10}$$

They represent partial oxidation, Boudouard gasification, and steam gasification. Their relative heat effects are important: (8) releases heat, whereas (9) and (10) consume heat. For each reaction, a kinetic coefficient and an external-diffusion coefficient are combined as resistances in series:

$$k_{\mathrm{eff},r} = \left(\frac{1}{k_{s,r}} + \frac{1}{k_d}\right)^{-1}, \qquad \dot{m}_{C,r} = k_{\mathrm{eff},r} A_p p_{\mathrm{ox},r}. \tag{11}$$

The diffusion closure is $k_d = C_{\mathrm{diff}}[(T_g + T_p)/2]^{0.75}/d_p$, with $C_{\mathrm{diff}} = 5.0 \times 10^{-12}$. The surface coefficient is

$$k_{s,r} = A_r T_p^{\beta_r} \exp\left(-\frac{E_r}{R_u T_p}\right). \tag{12}$$

The optimized BYU parameter set in Table III is taken from Table 6 of [2]. Those values are fitted for the reference coal and case; they should not be interpreted as universal coal constants. Particle external surface area is based on a fixed particle diameter and estimated particle count. Shrinking particle area, internal pore diffusion, ash effects, and evolving pore structure are omitted.

E. Global gas-phase chemistry

The model has four gas-phase global reactions:

$$\mathrm{CO} + \tfrac{1}{2}\mathrm{O_2} \rightarrow \mathrm{CO_2}, \tag{13}$$

$$\mathrm{H_2} + \tfrac{1}{2}\mathrm{O_2} \rightarrow \mathrm{H_2O}, \tag{14}$$

$$\mathrm{CO} + \mathrm{H_2O} \rightleftharpoons \mathrm{CO_2} + \mathrm{H_2}, \tag{15}$$

$$\mathrm{CH_4} + \mathrm{H_2O} \rightarrow \mathrm{CO} + 3\mathrm{H_2}. \tag{16}$$

Each forward rate has Arrhenius dependence multiplied by species concentrations raised to the orders in Table IV. The reversible water-gas-shift (WGS) rate is multiplied by $1 - Q/K_p$, with $Q = [\mathrm{CO_2}][\mathrm{H_2}]/([\mathrm{CO}][\mathrm{H_2O}])$ and $K_p = \exp(4577.8/T - 4.33)$. This compact mechanism omits intermediate hydrocarbon, tar, radical, and nitrogen chemistry; it has no turbulence-chemistry interaction closure.

The exact gas-phase constants and rate orders deserve particular caution. The code records these values but marks their provenance for verification. Consequently, the paper reports their implemented values for reproducibility and does not claim that the whole set has been independently validated against the selected reference mechanism. This uncertainty is relevant to the outlet residual oxygen and composition errors.

F. Radially coupled annular calculation

The radial calculation divides the vessel cross-section into N_r concentric annuli and the length into N_z axial stations. Each annulus carries a local vector of gas molar flows, char flow, raw-coal flow, and enthalpy flow. The inlet weights are constructed from the primary core and secondary annulus nozzle dimensions with a uniform-background fraction of 0.20; the discrete weights are normalized so integrated inlet stream rates match the case rates. This regularization avoids empty annuli, but it is not a measured radial velocity or species profile.

At each axial step, the local chemistry increment from the uncoupled reacting lane is applied to the previously mixed state. A backward-Euler finite-volume solve then diffuses each gas species between neighboring annuli. The same internal-face flux is entered with opposite signs in the adjacent control volumes, making the discrete radial species exchange conservative. Species enthalpy is carried with the diffusing molar flux, followed by an implicit radial heat-conduction step. The centerline has the axisymmetric symmetry condition; the wall has zero radial species and heat flux in the implemented closure.

The turbulent diffusivity is estimated from

$$D_t = \min(C_m U_b R,\ 0.20\ \mathrm{m^2\,s^{-1}}), \quad C_m = 0.05, \tag{17}$$

where U_b is the area-weighted bulk velocity and $R = D/2$. Species and heat diffusivities use $D_t/Sc_t + D_m$ and $D_t/Pr_t + D_m$, respectively, with $Sc_t = 0.70$, $Pr_t = 0.90$, $D_m = 10^{-5}\,\mathrm{m^2\,s^{-1}}$, and the same $0.20\,\mathrm{m^2\,s^{-1}}$ cap. The coefficient C_m, background inlet fraction, Schmidt number, Prandtl number, and cap form a reduced closure, not values inferred from experimental turbulence measurements.

Operator splitting is a computational choice. Chemistry increments are generated from independently solved axial lanes, then applied to a radially exchanged state at the next station. If the mixed state cannot supply a consumed trace species, the reaction increment is scaled to keep species nonnegative. This ensures robust local bookkeeping but means local chemistry does not continuously feed back into the same axial increment. No radial momentum equation updates local velocity in response to radial mass exchange.

V. Numerical Method and Verification

A. Integration and grid

The cross-section-averaged axial calculation integrates the species and solid states with LSODA, a stiff-capable variable-step integrator, using relative tolerance 10^{-6}, absolute tolerance 10^{-12}, and maximum step $L/200$. Temperature recovery uses a bracketed Brent root solve. The radial grid uses $N_r = 120$ annuli and $N_z = 600$ axial stations for the reported final calculation. The axial spacing is $L/(N_z - 1)$, and annular areas use exact differences of circular face areas. Radial species and thermal transport are solved with tridiagonal backward-Euler systems.

TABLE III
OPTIMIZED CHAR KINETIC PARAMETERS USED IN THE CODE. E IS SHOWN AFTER CONVERSION FROM J kmol^{-1} TO J mol^{-1}.

Reaction	A	E (J/mol)	β	Oxidizer
$C + 0.5O_2 \rightarrow CO$	1.298×10^{-3}	1.324×10^{5}	1.233	O_2
$C + CO_2 \rightarrow 2CO$	6.60×10^{-2}	1.385×10^{5}	-0.263	CO_2
$C + H_2O \rightarrow CO + H_2$	1.877×10^{-3}	1.588×10^{5}	-0.124	H_2O

TABLE IV
GLOBAL GAS-PHASE RATE SET AS IMPLEMENTED. VALUES ARE CODE INPUTS; THEIR COMPLETE SOURCE-TO-PARAMETER PROVENANCE IS NOT ESTABLISHED IN THE PROJECT FILES. THEY ARE NOT ASSERTED TO BE THE FULL MECHANISM OR PARAMETER SET OF [2]. CONCENTRATIONS ARE EVALUATED IN kmol m^{-3}.

Reaction	**Rate orders**	$\boldsymbol{A}$	$\boldsymbol{E}$ (J/mol)	**Heat of reaction used in enthalpy bookkeeping**
$CO + 0.5O_2 \rightarrow CO_2$	$[CO]^1[O_2]^{0.25}[H_2O]^{0.5}$	2.239×10^{12}	1.70×10^5	-282.99 kJ/mol CO
$H_2 + 0.5O_2 \rightarrow H_2O$	$[H_2]^1[O_2]^{0.5}$	1.00×10^{14}	1.70×10^5	-241.83 kJ/mol H2
$CO + H_2O \rightleftharpoons CO_2 + H_2$	$[CO]^1[H_2O]^1$	2.75×10^9	8.40×10^4	-41.20 kJ/mol forward reaction
$CH_4 + H_2O \rightarrow CO + 3H_2$	$[CH_4]^1[H_2O]^1$	3.00×10^8	1.26×10^5	$+206.00$ kJ/mol CH4

B. Balance diagnostic

At the final mesh, the maximum absolute tracked-element-flow discrepancy is $2.0549 \times 10^{-6}\%$. The diagnostic accounts for gas species, solid char, and residual raw coal and checks C, H, O, and N from inlet to outlet. This very small numerical residual is evidence that the implemented allocation and transport steps conserve these tracked elements. It is not evidence that the chosen kinetic parameters or closures represent the real reactor, nor does it include sulfur.

C. Grid refinement

Table V compares four meshes. Dry-gas fractions in this table use total dry gas as denominator, including argon, nitrogen, and residual oxygen; this is distinct from the inert-free reporting basis used later. Refinement from 80×400 to 120×600 changes area-average outlet temperature by 9.08 K, carbon conversion by -1.28 percentage points, and the listed total-dry CO, H2, and CO2 fractions by -0.65, -0.74, and $+0.80$ percentage points.

The trend suggests decreasing changes in outlet temperature at the two finest meshes, but carbon conversion and species continue to shift. This four-mesh sequence is a practical resolution check, not a formal order-of-accuracy study. It does not test the radial mixing closure, the operator-splitting error, the chemistry, or the temperature-ceiling issue. No conclusion about physical accuracy follows from mesh stability alone.

VI. RESULTS

A. Axial development and spatial fields

Fig. 1 shows area-average axial species and temperature behavior. The profiles summarize the radial states and should not be interpreted as centerline curves. The radial model also predicts strong radial variation because the primary and secondary feeds enter different coaxial regions and mix downstream.

Across the cross-section, the area-average outlet values from the saved radial profile are $T = 3556.7$ K, CO $= 24.42\%$, $H_2 = 7.27\%$, $CO_2 = 11.95\%$, $CH_4 = 0.210\%$, $H_2O = 48.33\%$, $O_2 = 2.079\%$, Ar $= 5.195\%$, and $N_2 = 0.537\%$ on the model's wet mole-fraction basis. The area-average outlet gas still contains oxygen. On a total dry-gas basis, area-average O_2 is about 4.02%, indicating that oxygen is not fully consumed in the calculation. At least three components could contribute: the assumed radial inlet distribution and mixing, the simplified gas reaction set, and the absence of solved turbulent flow and recirculation. The present data do not isolate these causes.

B. Outlet composition and conversion comparison

For comparison with the dry-gas benchmark, the model's four dry, inert-free product species are renormalized to sum to 100%, excluding steam, argon, nitrogen, and residual oxygen. This is a reporting normalization and does not mean that oxygen is physically absent from the outlet. Table VI contrasts the model with the measured BYU values summarized in the selected study. The clearest discrepancies are 7.56 percentage points excess CO, 8.02 points deficient H2, and 28.48 points lower carbon conversion. CO2 is within 0.61 points under this outlet normalization, but pointwise validation in Section VI-C shows that the axial and radial shapes still differ.

For product species $i \in \{\mathrm{CO}, \mathrm{H_2}, \mathrm{CO_2}, \mathrm{CH_4}\}$, the reporting basis is

$$x_{i,\mathrm{dry,inert\text{–}free}} = 100 \frac{F_{i,\mathrm{out}}}{\sum_{j\in\{\mathrm{CO,H_2,CO_2,CH_4}\}} F_{j,\mathrm{out}}}. \quad (18)$$

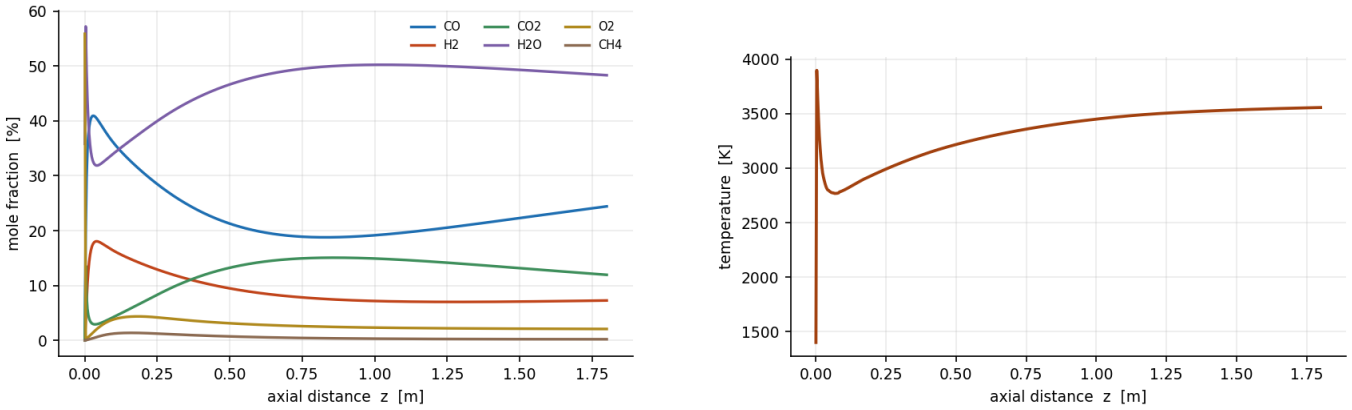
Carbon conversion is calculated from carbon remaining in the outlet char and raw coal:

$$X_C = 1 - \frac{m_{\mathrm{char,out}} + y_C m_{\mathrm{raw,out}}}{y_C m_{\mathrm{DAF,in}}}, \quad (19)$$

where y_C is the coal's dry-ash-free carbon mass fraction. This metric counts all unconverted char, including the surplus-carbon surrogate formed during volatile allocation.

TABLE V
MESH SENSITIVITY OF THE RADIAL CALCULATION. SPECIES VALUES ARE OUTLET DRY VOL% ON A TOTAL DRY-GAS BASIS, INCLUDING INERTS AND RESIDUAL OXYGEN. PEAK TEMPERATURE IS THE RECOVERED FIELD MAXIMUM AND IS CAPPED AT THE SOLVER'S UPPER BRACKET.

N_r	N_z	$T_{\mathrm{out,area}}$ (K)	X_C (%)	CO (%)	H2 (%)	CO2 (%)	Max. C/H/O/N residual (%)
20	100	3447.9	62.47	55.95	17.25	14.62	4.69×10^{-7}
40	200	3530.3	57.41	53.46	14.87	18.73	6.28×10^{-7}
80	400	3547.6	54.80	50.71	14.98	19.86	1.19×10^{-6}
120	600	3556.7	53.52	50.07	14.24	20.66	2.05×10^{-6}



(a) Area-average species profiles

(b) Area-average temperature profile

Fig. 1. Area-weighted axial averages from the radially coupled calculation. The hot start and temperature upper-bracket behavior are described in Sections VI-A and VII.

TABLE VI
DRY INERT-FREE OUTLET COMPARISON. THE MODEL FRACTIONS ARE NORMALIZED OVER CO, H2, CO2, AND CH4 ONLY.

Quantity	Model	BYU data	Difference
CO (vol%)	58.66	51.1	+7.56 pp
H2 (vol%)	16.68	24.7	−8.02 pp
CO2 (vol%)	24.21	23.6	+0.61 pp
CH4 (vol%)	0.45	0.3	+0.15 pp
Carbon conversion (%)	53.52	82	−28.48 pp

The cold-gas-efficiency diagnostic is 25.95%, calculated as

$$\mathrm{CGE} = \frac{\sum_{i\in\{\mathrm{H_2,CO,CH_4}\}} F_{i,\mathrm{out}}\mathrm{LHV}_i}{m_{\mathrm{DAF,in}}\mathrm{HHV}_{\mathrm{daf}}}. \tag{20}$$

Its numerator uses lower heating values of 10.8, 12.6, and 35.8 MJ/Nm^3 for H2, CO, and CH4, respectively, with a reference molar volume of 22.4 L/mol; the fuel denominator is based on the dry-ash-free coal HHV estimated by (4). It is reported as a code-defined diagnostic, not as an independently validated plant efficiency. Because the current chemistry leaves residual oxygen and substantially underpredicts conversion, this CGE should be read with those limitations in view.

C. Centerline profile comparison

The axial validation compares the innermost radial-cell center at $r = 0.00083$ m with approximate centerline experimental markers digitized from Fig. 9 of [2]. Marker coordinates are not raw tabulated measurements: the project digitization has estimated uncertainties of approximately ± 0.3 mole-fraction percentage point and ± 0.01 m in axial position. The error definition is the mean or maximum absolute composition difference in percentage points at the digitized axial samples:

$$e_{\mathrm{abs},i} = |x_{i,\mathrm{model}} - x_{i,\mathrm{exp}}| \times 100. \tag{21}$$

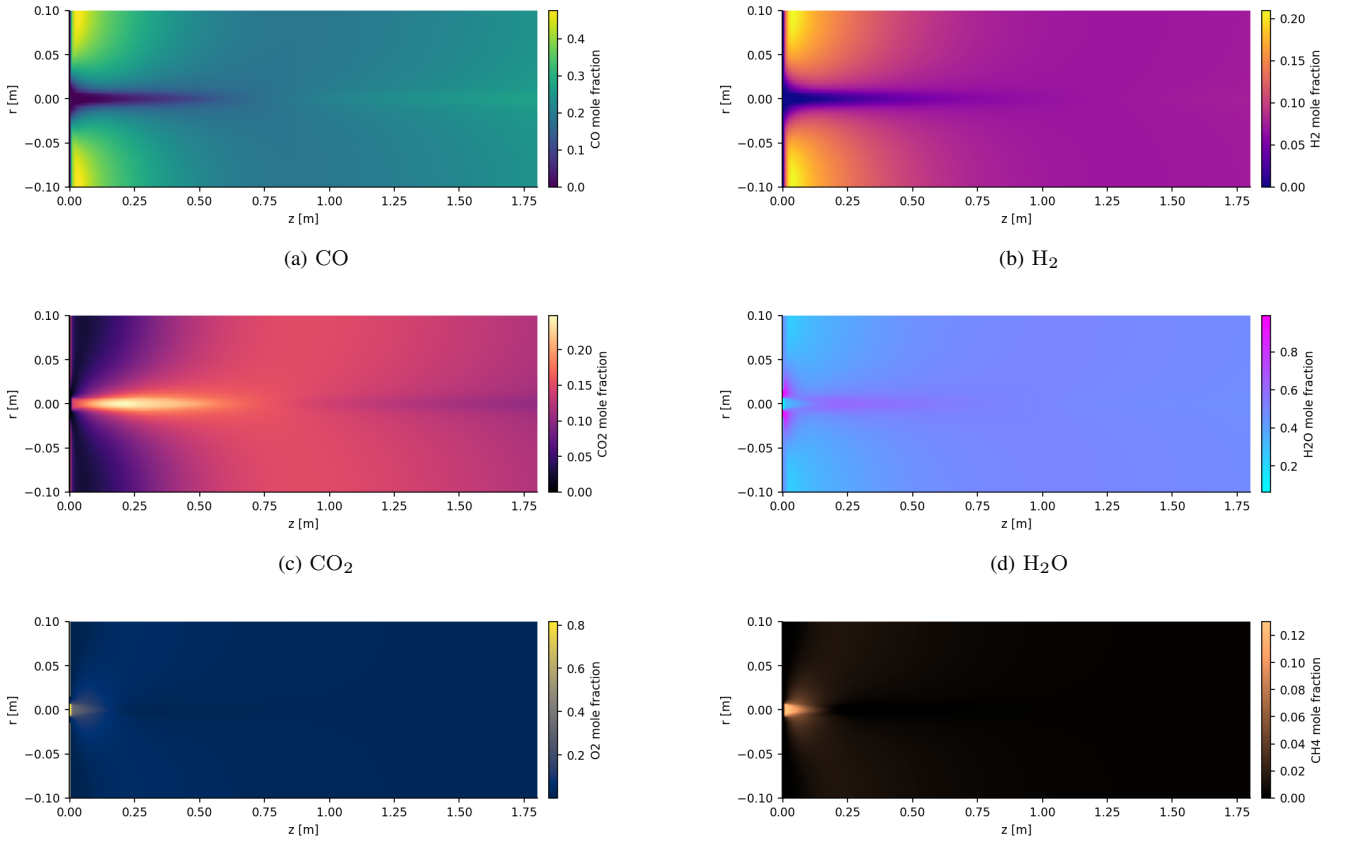
Table VII compares the calculated errors with the error values reported by the paper for its own CFD model. The latter are a contextual benchmark and not a re-analysis of our model using the paper's original simulation fields.

The centerline errors for the three scored species exceed the reference model's reported mean and maximum errors. Therefore, the current radial model does not reproduce the measured axial profiles at the reference-model level of agreement. The differences are consistent with the simplified flow, mixing, particle, thermal, and gas-chemistry closures, but the current analysis cannot attribute an error fraction to any one closure.

D. Radial profile comparison

Radial measurements at eight axial stations ($z = 0.13, 0.20, 0.28, 0.34, 0.51, 0.81, 1.12,$ and 1.73 m) are digitized from Figs. 12–14 of [2]. The source plot's orange experimental markers are retained. Their approximate digitization uncertainty is ± 0.001 m in radius and ± 0.5 percentage point in mole fraction. Pooled mean and maximum absolute errors are shown in Table VIII; H2O is excluded for the same balance-derived-data reason as in the axial comparison.

The radial pooled means are smaller than some of the centerline errors, but maximum local differences remain as

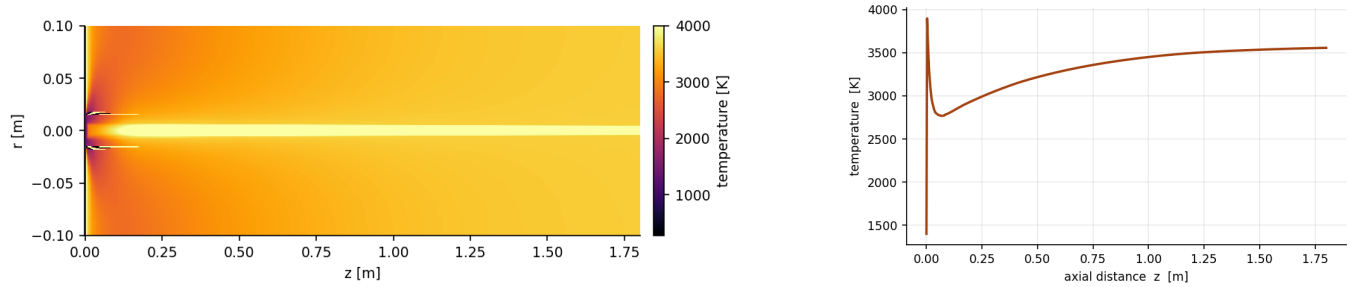


(a) CO (b) H_2

(c) CO_2 (d) H_2O

(e) O_2 (f) CH_4

Fig. 2. Computed radial-axial mole-fraction fields. These are model fields from the reduced annular calculation, not experimental contour measurements.



(a) Temperature field (b) Area-average axial temperature

Fig. 3. Temperature distribution from the model. A recovered value of 4000 K is the upper enthalpy-root bracket; the reported maximum is clipped and is not treated as a measured or converged peak temperature.

high as 24.22 percentage points for CO and 20.51 points for CO2. A pooled average can hide strong local disagreements, so these values do not justify a general claim of agreement. Digitization uncertainty is also not a substitute for the original raw measurement uncertainty.

E. Particle-size sensitivity

The independent particle-size sweep uses the cross-section-averaged axial calculation. It varies Rosin–Rammler scale diameter over 20, 36, and 60 μm and spread parameter over 0.7, 1.033, and 1.5, with six mass-weighted size bins; a 36-μm monodisperse case is included. All cases return approximately 100% carbon conversion, 71.65% cold-gas efficiency, and nearly identical composition. Table IX summarizes representative results.

The sweep is inconclusive for particle-size influence because conversion has saturated at 100%. The reference study reports 83% carbon conversion for a 36-μm monodisperse case and 77% for the full 1–85-μm distribution, versus 82% measured [2]. Those values are reported here only as outcomes of the reference study; they are not predictions reproduced by this reduced model. A useful follow-on study would lower the

TABLE VII
NEAR-AXIS CENTERLINE ERROR COMPARISON. ERRORS ARE ABSOLUTE MOLE-FRACTION DIFFERENCES IN PERCENTAGE POINTS. THE SELECTED STUDY'S REPORTED VALUES REFER TO ITS OWN MODEL.

Species	n	Present mean (pp)	Present max (pp)	Reference mean (pp)	Reference max (pp)	Scored?
CO	7	12.811	25.722	4.82	18.48	Yes
H2	6	10.498	13.027	5.47	7.95	Yes
CO2	7	7.268	14.458	4.72	10.15	Yes
H2O	6	25.541	31.337	9.58	20.22	No; derived and uncertain

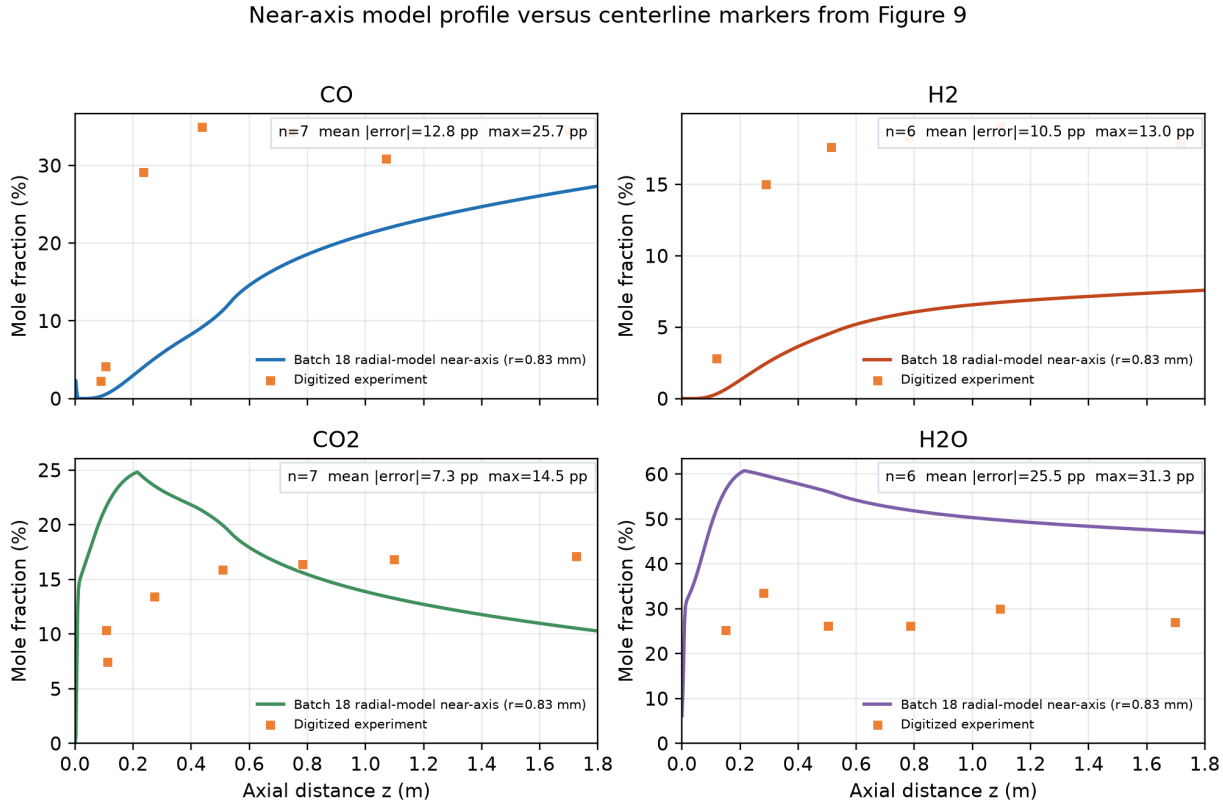


Fig. 4. Near-axis axial model profiles and digitized centerline markers. The reported H2O marker series is shown for context but not scored because the source study derives H2O by elemental balance and assigns it approximately $\pm 14\%$ uncertainty.

TABLE VIII
POOLED RADIAL-PROFILE VALIDATION ERROR OVER ALL SCORED STATIONS.

Species	n	Mean absolute error (pp)	Maximum (pp)
CO	34	8.541	24.223
H2	36	7.023	15.327
CO2	35	6.737	20.507

TABLE IX
REPRESENTATIVE PARTICLE-SIZE-SWEEP OUTPUTS FROM THE CROSS-SECTION-AVERAGED CALCULATION.

Distribution	Scale (μm)	X_C (%)	CGE (%)	H2/CO
Rosin–Rammler	20	100.00	71.65	0.451
Rosin–Rammler	36	100.00	71.65	0.451
Rosin–Rammler	60	100.00	71.65	0.451
Monodisperse	36	100.00	71.65	0.451

excessive burnout tendency or extend the sensitivity to particle density, char kinetics, and residence time and then verify that a physically meaningful unsaturated regime is obtained.

VII. ENERGY INITIALIZATION AND INTERPRETATION

The 1400 K hot-start is applied to avoid a cold, non-igniting plug-flow solution. A one-dimensional reacting stream without recirculation or radiation feedback has no mechanism to establish a flame from the cold inlet streams at the outset. In the implementation, the hot-start raises the enthalpy reference for the mixture before axial integration. Its equivalent added energy is approximately 19.35 kW, about 9.75% of coal HHV input. This is an artificial energy input used to initiate chemistry; it is not generated by the coal and is not a solved burner heat-transfer process.

This patch affects temperature-dependent devolatilization and reaction rates from the beginning of the calculation. It can

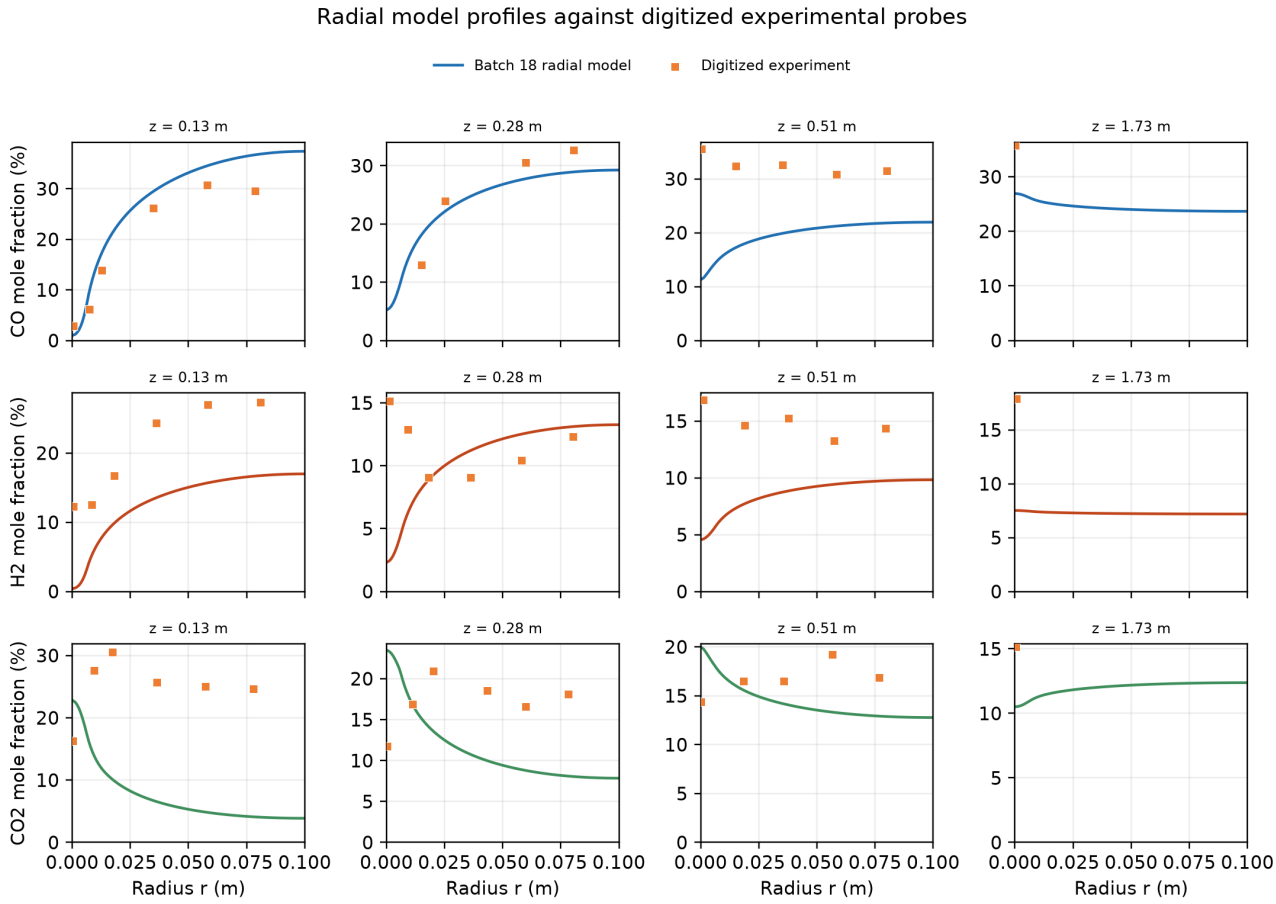


Fig. 5. Radial concentration profiles at representative axial stations, compared with digitized experimental markers. The complete error calculation includes all eight measurement stations, even though the plot displays a subset.

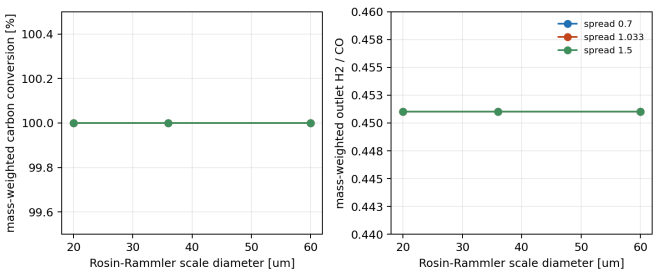


Fig. 6. Particle-size sweep. The flat result diagnoses complete-burnout saturation in this axial calculation; it does not demonstrate that particle size has no physical effect.

therefore alter the subsequent thermal and chemical solution, not merely shift the temperature plot. A predictive version should replace it with a physical ignition/burner boundary condition or a resolved recirculation/radiation model. The current temperature field also reaches the 4000 K enthalpy-root bracket. Since the bracket itself sets this ceiling, no physical interpretation should be assigned to the clipped peak.

VIII. Discussion

A. What the model establishes

The source code produces two complementary spatial calculations within one reproducible workflow. The cross-section-averaged plug-flow result supports the simpler axial baseline and particle-size sweep; the radially coupled result supplies the contours and the centerline and radial profile comparisons. Its strongest verified numerical property is conservation of C/H/O/N within the implemented state and transport formulation. Grid refinement has a modest final-step effect on outlet area-average temperature, but composition and carbon conversion remain more sensitive than the temperature alone.

B. What the model does not establish

The $2.05 \times 10^{-6}\%$ tracked-element residual establishes arithmetic consistency for C/H/O/N, not physical correctness. The final outlet comparison shows a 28.48-point conversion deficit and an 8.02-point H2 deficit. The model also leaves approximately 4.02 vol% dry area-average oxygen and reaches a temperature ceiling. Axial and radial errors show local mismatches. Together, these observations prevent a claim that the model is a validated predictor of reactor performance.

Several limitations may contribute, but their individual effects have not been isolated:

- no momentum equation, turbulence closure, recirculation, or residence-time distribution;
- assumed coaxial inlet weights and radial turbulent diffusivity rather than solved jet mixing;
- no particle trajectories, particle slip, particle radial dispersion, shrinking diameter, or separate particle temperature;
- no radiative heat transfer or wall heat loss, and a 1400 K hot-start energy addition;
- simplified volatile allocation without explicit tar and a sulfur-free product set;
- a four-reaction gas mechanism whose exact parameter provenance is incomplete in the project record; and
- numerical clipping at the 4000 K enthalpy bracket and a mixing/chemistry operator split.

C. Why a reduced implementation was used

A reduced formulation allows the full chain from feed analysis to species and energy balances, chemistry, radial transport, numerical integration, and validation to remain inspectable and demonstrable. The same reduction creates the limitations above. Reproducing the reference CFD would require a gas-flow solver, turbulence model, Lagrangian particles and dispersion, radiation, tar-aware devolatilization, validated gas chemistry, and coupled heterogeneous surface chemistry. These components are coupled and require consistent boundary conditions, model parameters, and numerical methods. The present implementation does not claim to substitute for that solver. It uses common benchmark inputs and published char/devolatilization parameter subsets to study and expose how a transparent reduced closure behaves.

IX. Conclusions

This work documented and evaluated a reduced-order BYU entrained-flow gasifier model. The model uses the published geometry, coal, stream conditions, competing two-step devolatilization parameters, and optimized char-kinetic parameters, while a conservative finite-volume radial-diffusion closure extends reacting axial lanes across the reactor radius. The model supplies feed tables, axial species profiles, radial-axial species and temperature fields, grid and particle-size studies, and comparisons with digitized axial and radial experimental data.

At 120×600, the reported outlet area-average temperature is 3556.7 K and carbon conversion is 53.52%. Dry inert-free CO/H2/CO2/CH4 are 58.66/16.68/24.21/0.45 vol%, compared with 51.1/24.7/23.6/0.3 vol% summarized for the BYU measurements. Carbon conversion is 28.48 percentage points below the reported 82% experiment. The dry outlet still contains about 4.02 vol% area-average O2, and the local maximum temperature is limited by the 4000 K root bracket. Centerline mean absolute errors for CO/H2/CO2 are 12.81/10.50/7.27 points; pooled radial mean errors are 8.54/7.02/6.74 points. H2O comparison is shown only as context because the experimental value is derived and carries large uncertainty.

The mesh study shows small changes in the finest-grid outlet temperature but does not eliminate sensitivity in conversion and composition or establish accuracy. The particle-size sweep is inconclusive because the axial calculation burns out all cases. The correct engineering conclusion is therefore that the project achieved a transparent, balance-checked reduced solver and a complete comparison workflow, while the current model does not reproduce all measurements and needs physical and chemical improvements before predictive use.

X. Recommended Model Improvements

The following changes follow directly from the diagnosed issues:

1) verify and document every gas-phase pre-exponential factor, activation energy, rate order, concentration unit, and reverse-rate convention against a primary source; then rerun the outlet and profile comparisons;
2) remove the artificial hot-start by implementing a physical burner or recirculation boundary condition and account for wall radiation and heat losses;
3) solve gas-phase momentum and turbulence, including recirculation and a consistent chemistry–turbulence interaction closure;
4) transport particles with separate particle temperature, slip, size-dependent trajectories, dispersion, shrinking surface area, and residence time;
5) replace the ad hoc volatile split and surplus-carbon-to-char approximation with an explicit tar-aware devolatilization and cracking model;
6) include sulfur-bearing products or explicitly expand the case's tracked element set; and
7) conduct mesh, axial-step, radial-mixing-parameter, hot-start, particle-density, and kinetic-parameter sensitivity studies with unsaturated conversion and un-clipped temperature outcomes.

These are model-development recommendations, not results already achieved in the present simulations.

XI. Reproducibility and Project Artifacts

A. Reproduction

The project source tree contains the case definition in `case/byu_case.yaml`, model modules in `src/gasifier/`, run scripts in `scripts/`, validation data and provenance in `data/validation/`, and computed tables and plots in `report/`. The conda environment file and notebook kernel instructions accompany the live-demo notebook. The saved radial-profile table and headline metrics are under `report/`; grid and particle-size results are in `report/grid_independence.csv` and `report/particle_size_sensitivity.csv`. The validation errors are in `report/validation_errors.csv` and `report/validation_radial_errors.csv`.

The Overleaf package consists of this LaTeX source, `references.bib`, a `figures/` directory, and `README.md`. Upload the source ZIP as a new Overleaf project and set `Batch18_IEEE_Report.tex` as the main document. The local build uses the IEEEtran journal class and BibTeX.

B. Complete output set

The deliverables associated with the study are:

- the technical report as a compiled IEEE-style PDF and an editable Overleaf source package;
- the Python source code, BYU case input, saved numerical CSV outputs, plots, and validation-marker data;
- the live-demo Jupyter notebook, which explains the governing ideas and demonstrates an educational solver step by step;
- the class presentation PDF; and
- the traceable analysis evidence listed in Table II.

The live-demo notebook is a pedagogical demonstration and is not the numerical source of the 120-by-600 results reported in this paper. The result numbers here refer to the saved project radial calculation and associated CSV files.

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